Double Spacetime in Projective Geometric Algebra: Duality, Null Translations, and Motors

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Abstract

Double spacetime theory treats gravity as torsional mismatch between two Minkowski frames carried by each massive particle. Those frames live in the biquaternion algebra $\mathrm{Cl}(3,1)$. What they do not yet contain is translation. Projective geometric algebra $G(3,1,1)$ supplies the missing piece by adjoining a single null vector e_4. Translations become bivectors, on the same footing as boosts and rotations. The ten bivectors are a basis of the bivector space, which splits as the direct sum of three boosts, three rotations, and four translations. Under the commutator that space is the Poincaré algebra $\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$.

Four statements organise the construction. Duality $X \mapsto Xi$ is a complex structure on the Lorentz sector: every boost is the dual of a rotation, and the sign that separates $\mathfrak{so}(3,1)$ from $\mathfrak{so}(4)$ is $i^2 = -1$. The geometric object that unites torsion with translation is the ordered product $M = RT$ of a torsional rotor and a null translator, not a single Banach exponential. On a coordinate chart the exterior Schwarzschild geometry is a radial boost of that motor, and the finite-angle scalar J that measures mismatch in parameter space is not the teleparallel field density T. The direction of gravity is carried by the translations: the static observer's motion is a single boost of rate GM/r^2 about an axis displaced inward by the inverse hover acceleration, and gravity points to that axis. In the flat flow of that bivector a body released from rest accelerates toward the axis at rate $1/\ell$ and meets the light cone of the axis in proper time ℓ. Written for radial rapidity and integrated while the axis distance changes with radius, the same law is the radial fall of the exterior: the areal radius accelerates at the boost rate, and one integration recovers $(dr/d\tau)^2 = E^2 - A$. The axis reaches the observer at the horizon and recedes to infinity far away. Because the squared Killing norm has d'Alembertian $8J$, the sign of J decides whether bodies fall toward an axis or away from it. At the centre of a spherical mass the boost rate vanishes with the weight and the translation returns to its asymptotic value, so a clock deep underground runs fast relative to the surface. The continuum interior keeps a slowed translation where the weight has already vanished.

1 Introduction

The double spacetime theory refuses a shared continuum. Each massive particle carries a pair of Minkowski frames—the usual frame, in which we measure, and a dual frame, time-reversed relative to the first, with the same interval. Gravity is the torsion of that pair [6, 10]. The algebra that holds the pair is the sixteen-dimensional biquaternion algebra, canonically isomorphic to $\mathrm{Cl}(3,1)$. Boosts of the usual sector and rotations of the dual sector are generated by six bivectors. Their mismatch is a single algebraic object, and that object is enough to recover the exterior of a spherical mass without taking curvature as a primitive.

Six bivectors are not ten. The Lorentz algebra $\mathfrak{so}(3,1)$ accounts for orientation; it does not account for displacement. To move a particle from one event to another, the algebra must also generate translations. The natural home for that generator, in geometric algebra, is a null vector: a direction that squares to zero and therefore produces, when wedged with the four Minkowski generators, four bivectors that themselves square to zero. This is the standard convention of projective geometric algebra $G(3,1,1)$ [1, 2, 3, 4], going back to Selig's treatment of rigid motions in degenerate Clifford algebras [11]. Adjoining one null vector e_4 to $\mathrm{Cl}(3,1)$ lifts the double spacetime into a thirty-two-dimensional algebra whose bivector space is ten-dimensional. Those ten bivectors are three boosts, three rotations, and four translations.

The lift is not a bookkeeping device. Under the commutator the ten-dimensional span closes, the translations form an abelian ideal, and the Lorentz sector acts on that ideal by its vector representation. The result is the Poincaré algebra $\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$ realised inside $G(3,1,1)$ [12, 13]. Duality, already present in the parent theory as right multiplication by the pseudoscalar i, becomes a complex structure

on the Lorentz factor: boosts are dualised rotations, and $i^2 = -1$ forces the boost–boost sign that distinguishes Lorentz from Euclidean. Translations, by contrast, are sent out of grade two. Duality organises orientation; it does not organise displacement.

Once both sectors are in hand, a motor is an ordered product $M = RT$. The torsional factor R is the exponential of the six-dimensional mismatch; the translational factor T is linear, because a null bivector squares to zero. Unitarity $M\widetilde{M} = 1$ follows from the reversal-odd character of every bivector, which is why translations must be bivectors rather than vectors. When torsion and translation fail to commute, M is not the exponential of their sum. The geometric object is the product.

The same motor, restricted to a radial boost, recovers the diagonal Schwarzschild coframe. What it does not do is identify the parameter-space scalar J with the teleparallel torsion density T. One is a finite angle; the other is a derivative of the coframe. Confusing the two collapses a constant boost into a curved geometry, or a pure translation into a non-vanishing field.

Direction is a separate matter. A rapidity field says how strongly the coframe is boosted, not which way a released body falls. The answer lies with the translations that the chart reading leaves unused. The static observer moves by time translation, and in $G(3,1,1)$ the generator of that motion is a radial boost plus a time translation that do not commute. Their sum is a single boost about an axis displaced inward, and gravity points from the observer to that axis. The boost rate is the Newtonian field GM/r^2 without approximation, and the inverse distance to the axis is the hover acceleration. In the flat flow a body released from rest accelerates toward the axis at that rate, and the same distance is the proper time in which it meets the light cone of the axis. Once the law is written for the radial rapidity relative to the local hovering observer, the radius may change. The hyperbolic identity cancels the speed, the areal radius accelerates at GM/r^2, and the first integral is the radial Schwarzschild fall. The sign of J, which the parent theory reads as attraction or repulsion, becomes the sign of the curvature of the Killing norm. Inside a spherical mass the same bivector parts company with that tie: where the weight vanishes, at the centre, the translation returns to the asymptotic rate, and a buried clock runs fast relative to the surface. The continuum interior keeps a slowed translation there.

The paper follows that order. Section 2 adjoins e_4 and classifies the ten generators. Section 3 shows that duality determines the Lorentz algebra. Section 4 constructs the motor and the quadratic diagnostic $J^{(5)}$. Section 5 reads the Schwarzschild exterior as a radial boost. Section 6 places the direction of gravity at the axis of the static Killing bivector, and integrates the flat rapidity law, while the radius changes, into the radial Schwarzschild fall. Section 7 returns to what the algebra has settled and what it has not. Appendix B reads across that order. The non-commuting pair, the derivative ladder, and the sign of J are one construction. The algebraic identities of those sections are machine-checked in the Lean 4 companion `DstDiophantine`.

2 Adjoining a null direction

The parent theory identifies the four vector generators of $\mathrm{Cl}(3,1)$ with biquaternion elements

$$e_0 = j, \qquad e_1 = kI, \qquad e_2 = kJ, \qquad e_3 = kK,$$

of Minkowski signature $(-,+,+,+)$:

$$e_0^2 = -1, \qquad e_1^2 = e_2^2 = e_3^2 = +1, \qquad \{e_\mu, e_\nu\} = 0 \quad (\mu \neq \nu).$$

Projective geometric algebra $G(3,1,1)$ is obtained by adjoining one further generator e_4 with

$$e_4^2 = 0, \qquad \{e_4, e_\mu\} = 0 \quad (\mu = 0,1,2,3). \tag{1}$$

The full Clifford algebra is $2^5 = 32$-dimensional. The original sixteen-dimensional biquaternion algebra is the subalgebra generated by $\{e_0, e_1, e_2, e_3\}$.

The dual map of the parent theory is right multiplication by the biquaternionic pseudoscalar [9]

$$i \;=\; e_0e_1e_2e_3 \;=\; j\,(kI)(kJ)(kK).$$

Because e_4 anticommutes with each of the four Minkowski vectors, it commutes with their product:

$$e_4\, i \;=\; e_4\, e_0e_1e_2e_3 \;=\; (-1)^4\, e_0e_1e_2e_3\, e_4 \;=\; i\, e_4.$$

Duality therefore continues to act on $\mathrm{Cl}(3,1)$ exactly as before, while the new null direction is left intact. The lift does not disturb the time-reversing character of $X \mapsto Xi$.

The bivector space of $G(3,1,1)$ has dimension $\binom{5}{2} = 10$. Those ten elements fall into three geometric classes, distinguished by the square.

The three *hyperbolic* generators square to $+1$ and span the boosts of $\mathfrak{so}(3,1)$:

$$iI \;=\; e_0e_1, \qquad iJ \;=\; e_0e_2, \qquad iK \;=\; e_0e_3.$$

The three *cyclic* generators square to -1 and span the rotations of the same Lorentz copy:

$$I \;=\; e_3e_2, \qquad J \;=\; e_1e_3, \qquad K \;=\; e_2e_1.$$

The four *null* generators are the wedges of e_4 with the Minkowski basis,

$$\begin{aligned} N_0 &= e_4 \wedge e_0 = e_4\, j, & N_1 &= e_4 \wedge e_1 = e_4\,(kI), \\ N_2 &= e_4 \wedge e_2 = e_4\,(kJ), & N_3 &= e_4 \wedge e_3 = e_4\,(kK). \end{aligned}$$

From (1) one has, for every pair $\mu, \nu \in \{0,1,2,3\}$,

$$N_\mu N_\nu \;=\; (e_4e_\mu)(e_4e_\nu) \;=\; -\,e_4^2\, e_\mu e_\nu \;=\; 0. \tag{2}$$

Each N_μ squares to zero, and every mixed product vanishes as well. The four null bivectors commute; they will be the translations.

A light-like translation is not a constraint on the generators. A general Minkowski vector $L = ct\,e_0 + x\,e_1 + y\,e_2 + z\,e_3$ is null when $L^2 = 0$. The translation along L is the bivector $e_4 \wedge L$, a linear combination of the four N_μ, and it squares to zero by (2) whether or not L itself is null. Light-likeness lives on the four parameters $\lambda^\mu = (ct, x, y, z)$, as the cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$. The translation sector remains four-dimensional.

Table 1: The ten bivector generators of $G(3,1,1)$.

Class	Generators	Count	Square
Hyperbolic	$B_1^+ = iI,\ B_2^+ = iJ,\ B_3^+ = iK$	3	$+1$
Cyclic	$B_1^- = I,\ B_2^- = J,\ B_3^- = K$	3	-1
Null	$N_0 = e_4 j,\ N_1 = e_4(kI),\ N_2 = e_4(kJ),\ N_3 = e_4(kK)$	4	0

These ten elements are a basis of the bivector space. The three classes therefore span complementary subspaces, of dimensions three, three, and four, and every bivector is uniquely a boost plus a rotation plus a translation. The binomial count $\binom{5}{2} = 10$ is the dimension of this direct sum.

3 Duality determines the Lorentz algebra

The six hyperbolic and cyclic generators close among themselves. Same-axis boosts and rotations commute. Adjacent rotations reproduce the remaining rotation; adjacent boost–rotation pairs reproduce the remaining boost; adjacent boosts reproduce the remaining rotation with a reversed sign:

$$\begin{aligned} [B_a^-, B_{a+1}^-] &= 2B_{a+2}^-, \\ [B_a^+, B_{a+1}^-] &= 2B_{a+2}^+, \\ [B_a^+, B_{a+1}^+] &= -2B_{a+2}^-. \end{aligned}$$

Indices run cyclically over the three axes. These are the structure constants of $\mathfrak{so}(3,1)$ in the rotation–boost presentation $[J,J] \sim J$, $[J,K] \sim K$, $[K,K] \sim -J$. The minus sign in the last line is what separates the Lorentz algebra from compact $\mathfrak{so}(4)$, where boosts would bracket to $+J$.

The four null generators are moved among themselves by that Lorentz action. A radial boost exchanges time-translation with the spatial translation along its own axis and fixes the other two; a rotation turns the two translations in its plane into each other and fixes the rest. Representative brackets are

$$\begin{aligned} [B_0^+, N_0] &= 2N_1, & [B_0^+, N_1] &= 2N_0, & [B_0^+, N_2] &= 0, \\ [B_0^-, N_2] &= 2N_3, & [B_0^-, N_3] &= -2N_2, & [B_0^-, N_0] &= 0. \end{aligned}$$

Every such bracket remains null. Null generators commute with one another, by (2).

Write $\mathfrak{l}$ for the six-dimensional span of boosts and rotations, and $\mathfrak{n}$ for the four-dimensional null span. Their sum is the direct sum just obtained, and it is the whole bivector space. Bilinearity of the commutator lifts the generator identities to

$$[\mathfrak{l},\mathfrak{l}] \subseteq \mathfrak{l}, \qquad [\mathfrak{l},\mathfrak{n}] \subseteq \mathfrak{n}, \qquad [\mathfrak{n},\mathfrak{n}] = 0.$$

Thus $\mathfrak{l}$ and $\mathfrak{p} := \mathfrak{l} \oplus \mathfrak{n}$ are Lie subalgebras of $G(3,1,1)$, and $\mathfrak{n}$ is an abelian ideal of $\mathfrak{p}$. The quotient $\mathfrak{p}/\mathfrak{n}$ is $\mathfrak{l}$, and $\mathfrak{l}$ acts on $\mathfrak{n}$ by the vector representation just recorded. This is the semidirect product $\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$ as an algebraic fact about the ten bivectors. The action is faithful in the sense that motors require: $[B_0^+, N_0] = 2N_1 \neq 0$, so torsion and translation do not commute in general.

The internal structure of $\mathfrak{l}$ is duality. The map $X \mapsto Xi$ squares to -1, because $i^2 = -1$. The pseudoscalar i anticommutes with each vector e_μ and therefore commutes with every product of two of them: i is central on $\mathfrak{l}$. Together these say that right multiplication by i is a complex structure on the Lorentz sector, and that $\mathfrak{l}$ is a module over $\mathbb{R}[i] \cong \mathbb{C}$.

On the generators the complex structure is exact:

$$B_a^- \, i \;=\; B_a^+, \qquad B_a^+ \, i \;=\; -B_a^- \qquad (a = 0,1,2).$$

Every boost is the dual of the corresponding rotation. If $\mathfrak{r}$ is the three-dimensional span of the rotation generators, the boost sector is $\mathfrak{r}\,i$, and

$$\mathfrak{l} \;=\; \mathfrak{r} \;\oplus\; \mathfrak{r}\,i.$$

The Lorentz algebra is the complexification of the rotation algebra. This is the classical isomorphism $\mathfrak{so}(3,1) \cong \mathfrak{so}(3) \otimes \mathbb{C} \cong \mathfrak{sl}(2,\mathbb{C})$, written inside $G(3,1,1)$ as the statement that boosts are dualised rotations.

The commutator respects the structure. Since i commutes with every element of $\mathfrak{l}$,

$$[Xi, Y] \;=\; [X,Y]\,i \;=\; [X, Yi], \qquad [Xi, Yi] \;=\; [X,Y]\,i^2 \;=\; -[X,Y] \qquad (X, Y \in \mathfrak{l}).$$

The bracket is complex-bilinear. Consequently the entire Lorentz table is generated by the rotation table alone. From $[B_a^-, B_{a+1}^-] = 2B_{a+2}^-$ one obtains

$$[B_a^+, B_{a+1}^-] \;=\; [B_a^-, B_{a+1}^-]\,i \;=\; 2B_{a+2}^+, \qquad [B_a^+, B_{a+1}^+] \;=\; -[B_a^-, B_{a+1}^-] \;=\; -2B_{a+2}^-.$$

The minus sign in the boost–boost bracket is $i^2 = -1$. In Euclidean signature the four-dimensional pseudoscalar squares to $+1$, the same computation yields $[K,K] \sim +J$, and the complexification collapses to compact $\mathfrak{so}(4) \cong \mathfrak{so}(3) \oplus \mathfrak{so}(3)$. The Lorentz signature of the double spacetime is encoded in the duality map itself.

The null sector behaves differently. Because e_4 commutes with i,

$$N_\mu \, i \;=\; (e_4 e_\mu)\, i \;=\; e_4\,(e_\mu\, i),$$

and $e_\mu i$ is grade three in $\mathrm{Cl}(3,1)$, so $N_\mu i$ is grade four in $G(3,1,1)$. Duality sends each translation out of the bivector grade. The null ideal is not dual-closed. The complex structure belongs to the Lorentz factor of $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{n}$, not to the Poincaré span as a whole.

4 Motors

The extended mismatch is the sum of a torsional bivector and a translational bivector,

$$\Omega^{(5)}_{\mathrm{biv}} = \underbrace{\sum_{a=1}^{3} \Big(\frac{\alpha_a}{2}\, B_a^+ + \frac{\beta_a}{2}\, B_a^- \Big)}_{\Omega_{\mathrm{torsion}}} + \underbrace{\sum_{\mu=0}^{3} \frac{\lambda^\mu}{2}\, N_\mu}_{\Omega_{\mathrm{trans}}}. \tag{3}$$

The six coefficients α_a, β_a are the original mismatch of usual and dual sectors. The four coefficients λ^μ are a Lorentz four-vector of translational mismatch. Light-like translations are the cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$.

Because every mixed product $N_\mu N_\nu$ vanishes, the translational generator is nilpotent:

$$\Omega^2_{\mathrm{trans}} \;=\; \frac{1}{4} \sum_{\mu,\nu} \lambda^\mu \lambda^\nu \, N_\mu N_\nu \;=\; 0.$$

Its exponential therefore truncates at first order,

$$\exp(\Omega_{\text{trans}}) \;=\; 1 + \Omega_{\text{trans}}.$$

If $T(p) = 1 + \Omega_{\text{trans}}(p)$ and $T(q) = 1 + \Omega_{\text{trans}}(q)$, the product is $T(p)T(q) = T(p+q)$. Translators form an abelian group under multiplication. A translation is a pure shift; it introduces neither curvature nor higher-order terms.

The Lorentz span has nilpotents of its own. Off a common axis the balanced sum $B_0^+ + B_1^-$ lies in that span, meets the null ideal only at zero, and squares to zero, so its exponential truncates to the null rotation $1 + t(B_0^+ + B_1^-)$. The rotor fixes the null line $e_0 - e_3$ and the transverse axis e_2, and it shears

$$e_1 \mapsto e_1 + 2t(e_0 - e_3), \qquad e_0 + e_3 \mapsto (e_0 + e_3) + 4t\, e_1 + 4t^2(e_0 - e_3).$$

These truncations multiply by adding the parameter, and the opposite parameter is the reverse, hence the inverse. The shear preserves the Minkowski product. For a nonzero parameter the fixed vectors are the span of $e_0 - e_3$ and e_2, and the two directions are orthogonal. Since e_4 commutes with the Lorentz span, a translation is sent to the translation with the sheared coefficients, and a lightlike translation remains lightlike. A translator $1 + sN_1$, of the same truncated shape, sends the hyperplane e_1 to $e_1 + se_4$.

The geometric motor is the ordered product of the two exponentials,

$$M \;:=\; RT, \qquad R \;:=\; \exp(\Omega_{\text{torsion}}), \qquad T \;:=\; 1 + \Omega_{\text{trans}}.$$

The order matters. Cross-commutators $[B_a^\pm, N_\mu]$ are generally nonzero, so Ω_{torsion} and Ω_{trans} need not commute, and M need not equal $\exp(\Omega_{\text{biv}}^{(5)})$. When they do commute, the two expressions coincide. When they do not—a radial boost against a time translation is the simplest example—the geometric object remains the product RT. A pure radial boost conjugates translators to translators: the sandwich $R(\varphi)\,T(p)\,\widetilde{R(\varphi)}$ is the translator whose coefficients are the Lorentz action of rapidity φ on the boost plane.

Unitarity of M is a consequence of grade. Reversal sends every bivector to its negative, $\widetilde{B_a^\pm} = -B_a^\pm$ and $\widetilde{N_\mu} = -N_\mu$. Had translations been taken as grade-one null vectors, they would have been reversal-even, and the argument below would fail. That is why projective geometric algebra generates translations by null *bivectors* [1, 2, 4]. For the rotor one has the standard cancellation

$$R\,\widetilde{R} \;=\; \exp(\Omega_{\text{torsion}})\,\exp(-\Omega_{\text{torsion}}) \;=\; 1.$$

For the translator, reversal-oddness together with nilpotence gives

$$T\,\widetilde{T} \;=\; \big(1 + \Omega_{\text{trans}}\big)\big(1 - \Omega_{\text{trans}}\big) \;=\; 1 - \Omega_{\text{trans}}^2 \;=\; 1.$$

Reversal is an antihomomorphism, so

$$M\,\widetilde{M} \;=\; RT\,\widetilde{T}\,\widetilde{R} \;=\; R\,\widetilde{R} \;=\; 1.$$

The motor is unitary whether or not the factors commute.

Duality acts on the torsional parameters as a quarter-turn. Using $B_a^- i = B_a^+$ and $B_a^+ i = -B_a^-$,

$$\Omega_{\text{torsion}}(\alpha, \beta)\, i \;=\; \sum_a \Big(\frac{\beta_a}{2}\, B_a^+ - \frac{\alpha_a}{2}\, B_a^- \Big) \;=\; \Omega_{\text{torsion}}(\beta, -\alpha).$$

Applied twice, the map reverses all six coefficients, in agreement with $i^2 = -1$. Writing $z_a = \alpha_a + \mathrm{i}\beta_a$, duality is multiplication by $-\mathrm{i}$. The torsional scalar

$$J \;=\; \tfrac{1}{2} \sum_a (\alpha_a^2 - \beta_a^2) \;=\; \tfrac{1}{2}\,\mathrm{Re} \sum_a z_a^2$$

is the real part of the holomorphic quadratic; the unsigned mass $\frac{1}{2}\sum_a |z_a|^2$ is its Hermitian companion. Duality therefore reverses J and preserves the mass. The usual/dual swap of the parent theory, which exchanges α_a with β_a, differs from this quarter-turn only by the sign of the new rotation parameter; both reverse J. Translational parameters must be tracked separately, because duality sends each N_μ to grade four.

A quadratic diagnostic on the full ten-dimensional space cannot come from the Cartan–Killing form of $\mathfrak{iso}(3,1)$ alone. That form vanishes on translations [14, 15]. Extending the Killing quadratic of the Lorentz sector by the Minkowski metric on the translation coefficients gives a non-degenerate pairing

$$J^{(5)} \;:=\; \tfrac{1}{16}\, B_{\mathrm{Killing}}(\Omega_{\mathrm{torsion}}, \Omega_{\mathrm{torsion}}) \;+\; \tfrac{1}{2}\, \eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu}. \tag{4}$$

Under the half-angle expansion of (3), with $B(B_a^+, B_a^+) = +8$ and $B(B_a^-, B_a^-) = -8$, the first term equals $\frac{1}{8}\sum_a(\alpha_a^2 - \beta_a^2)$, which is one quarter of the discrete companion's $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$. The second term vanishes precisely on the light cone. The two contributions decouple because the Killing form of $\mathfrak{iso}(3,1)$ vanishes between the semisimple and translation generators. The translational contribution is unbounded unless λ^μ is constrained; the admissible ceiling $|J| \le 3\pi^2/8$ of the discrete companion applies to the torsion part alone.

One geometric reading of the same algebra is worth recording, although it is not needed for the motor calculus. Planes in $G(3,1,1)$ are three-blades. The antisymmetric part of their geometric product,

$$A \veebar B \;:=\; \tfrac{1}{2}(AB - BA),$$

takes values in grade two. On representative configurations, if P is a three-blade and D a dual-sector element aligned with the normal n of the plane of P, the bracket is a null bivector along that normal,

$$P \veebar D \;=\; k\,(e_4 \wedge n).$$

The dual spacetime then acts as a carrier of normals, and the bracket produces a translation along the normal. The appendix works one such example. A fully covariant calculus relating these brackets to $\Omega^{(5)}_{\mathrm{biv}}$ is not developed here.

5 Gravity on a chart

The parent theory aims to realise teleparallel gravity: geometry is carried by torsion rather than by a curvature tensor. In the projective extension the same idea is available on a fixed coordinate chart. Throughout we work with the Poincaré subalgebra of Section 3. Variational equivalence of the teleparallel scalar with the Einstein–Hilbert action on a general manifold is taken from the TEGR literature [16] and is not re-derived.

A unitary motor acts on the reference coframe by sandwiching. Fix the $\mathrm{Cl}(3,1)$ vector generators $\{\mathring{e}^a\}_{a=0}^3$ inside $G(3,1,1)$, and set

$$e^a \;:=\; \big\langle M\, \mathring{e}^a\, \widetilde{M} \big\rangle_1. \tag{5}$$

On a chart (x^μ) the component fields $e^a{}_\mu(x)$ induce the metric

$$g_{\mu\nu} \;=\; \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu, \tag{6}$$

with $\eta = \mathrm{diag}(-1,+1,+1,+1)$. When M is a pure Lorentz rotor, the sandwich is a local Lorentz action on the reference vectors. Null translations enter the full motor $M = RT$ but do not by themselves supply the classical Schwarzschild diagonal gauge; they remain available as translational mismatch measured by $J^{(5)}$. They do enter the motion of the static observer, and there they carry the direction of gravity (Section 6).

The Weitzenböck torsion of a coframe on the chart is the antisymmetrised derivative

$$T^a{}_{\mu\nu} \;=\; \partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu. \tag{7}$$

With inverse tetrad $e_a{}^\mu$ and spacetime indices lowered by $g_{\mu\nu}$, the teleparallel scalar is

$$T \;=\; \tfrac{1}{4}T_{\rho\mu\nu}T^{\rho\mu\nu} + \tfrac{1}{2}T_{\rho\mu\nu}T^{\nu\mu\rho} - T_\rho T^\rho, \tag{8}$$

where $T_\rho = T^\sigma{}_{\rho\sigma}$. No curvature tensor of a metric-compatible Levi-Civita connection is required: the teleparallel description stores geometry in $T^a{}_{\mu\nu}$.

The exterior Schwarzschild geometry is a radial boost of this construction. Let $r_s = 2GM/c^2 > 0$ and work in the region $r > r_s$. The diagonal coframe (units $c = 1$) is

$$\begin{aligned} e^0 &= \sqrt{1 - \frac{r_s}{r}}\, dt, \\ e^1 &= \big(1 - \frac{r_s}{r}\big)^{-1/2}\, dr, \\ e^2 &= r\, d\theta, \\ e^3 &= r \sin\theta\, d\phi. \end{aligned} \tag{9}$$

Set $A(r) := 1 - r_s/r$ and

$$\varphi(r) \;:=\; -\log\sqrt{A(r)} \;=\; -\tfrac{1}{2}\log A(r) \qquad (r > r_s). \tag{10}$$

Then $e^{-\varphi} = \sqrt{A}$ and $e^{\varphi} = A^{-1/2}$, so the (t, r) diagonal coefficients are the boost scale factors of the pure hyperbolic rotor generated by $B_1^+ = e_0 \wedge e_1$. The angular legs come from the spherical reference coframe. The corresponding torsional parameter is the pure boost $\Omega_{\text{torsion}} = (\varphi/2)B_1^+$, with vanishing null parameters in the diagonal gauge. The induced metric (6) is the Schwarzschild metric.

Three scalars now sit on the same chart, and they are not the same object. The finite-angle diagnostic

$$J(\varphi) \;=\; \tfrac{1}{2}\varphi(r)^2$$

measures the size of the boost in parameter space. The field seed

$$J_{\text{field}}(r) \;:=\; \tfrac{1}{2}\big(\varphi'(r)\big)^2, \tag{11}$$

with $\varphi' = -r_s/(2r^2A)$, measures the radial rate of that boost. The teleparallel scalar (8) is a quadratic in the derivatives of the coframe. On this exterior chart the vacuum condition $A + rA' = 1$ freezes it at the value of the flat spherical frame,

$$T \;=\; -\frac{2}{r^2},$$

which is the radial divergence $T = -r^{-2}\partial_r(r^2 B^r)$ for $B^r = 2/r$. The mass drops out of T and does not drop out of J. The identity $\partial_r\sqrt{A} = -\varphi'\sqrt{A}$ relates the boost torsion leg to φ', but it does not identify J or J_{field} with T.

A constant nonzero boost has $J \neq 0$ while a flat chart has $T = 0$. The cancellation persists when the rapidity varies. A boost of the inertial frame in one fixed plane, with rapidity an arbitrary function of the two coordinates in that plane, still induces the Minkowski metric, the (t, x) block has determinant one, and the teleparallel scalar remains zero. The torsion need not: its legs are $T^t{}_{tx} = \partial_t\varphi$ and $T^x{}_{xt} = \partial_x\varphi$. The finite angle $J = \frac{1}{2}\varphi^2$ is not recovered from T, nor from the vacuum equation, which holds for every such profile because the metric does. A pure spatial translation on Minkowski space has $J^{(5)} \neq 0$ and $T = 0$. Even the field seed fails to match $\frac{1}{2}T$ at a generic exterior point: at $(r_s, r) = (2, 4)$, for instance, the two numbers disagree. Finite angle, rate of angle, and coframe density live on different rungs of the derivative ladder. A dictionary for a general smooth motor field $M(x) = R(x)T(x)$ would have to be built from derivatives of $(\Omega_{\text{torsion}}, \lambda)$, not from finite angles, and would be expected to involve a total divergence. The pure-boost family shows that the dictionary is not a function of the finite angle. It is not constructed here.

Sandwiching with respect to the full degenerate metric of $G(3, 1, 1)$ is likewise left aside. The identification of M with $\exp(\Omega^{(5)}_{\text{biv}})$ holds when the factors commute and fails in general otherwise. The simplest failure is the static observer itself, to which we now turn.

6 The direction of gravity

The radial boost of Section 5 fixes the metric. It does not say which way a released body falls. The rapidity $\varphi(r)$ is a scalar, and a scalar acquires a direction only through something that moves across it. What moves is the static observer. Its motion is a time translation, and in $G(3, 1, 1)$ the generator of a motion is a bivector.

At an event a Killing field ξ is fixed by two data: its value, which is a translation, and its antisymmetric derivative $\nabla\xi$, which is a Lorentz generator [17]. The pair is an element of the Poincaré algebra, and by Section 3 that element is a bivector of $G(3, 1, 1)$. For the static field ∂_t of the exterior chart, in the orthonormal frame of the static observer, the value is $\nu\, e_0$ with

$$\nu \;=\; \sqrt{A} \;=\; e^{-\varphi},$$

and the only component of $\nabla\xi$ is a radial boost, whose rate is the derivative of ν along the proper radial distance s:

$$\kappa \;=\; \frac{d\nu}{ds} \;=\; \sqrt{A}\,\partial_r\sqrt{A} \;=\; \tfrac{1}{2}A' \;=\; \frac{r_s}{2r^2}.$$

The static Killing bivector is

$$K \;=\; \frac{\kappa}{2}\,B_1^+ \;+\; \frac{\nu}{2}\,N_0. \tag{12}$$

Its translation part is the redshift and its Lorentz part is the boost rate. With $r_s = 2GM$ the boost rate is $\kappa = GM/r^2$ at every radius, without a weak-field expansion: the Newtonian field is the Lorentz part of the static Killing bivector. Gauss's law holds for it without correction. The product $r^2\kappa = r_s/2$ does not depend on r, so the flux $4\pi r^2\kappa = 4\pi GM$ through every sphere is the same. This is Komar's mass of the exterior [18].

The pair in (12) is the non-commuting example of Section 4, a radial boost against a time translation, and it is not their product. Conjugation by a translator is exact at first order,

$$T\,X\,\widetilde{T} \;=\; X + [\Omega_{\text{trans}}, X],$$

because a sandwich between two null bivectors vanishes. Applied to the pure boost it gives

$$K \;=\; T(-\ell e_1)\left(\frac{\kappa}{2}B_1^+\right)\widetilde{T}(-\ell e_1), \qquad \ell \;=\; \frac{\nu}{\kappa}. \tag{13}$$

The static Killing bivector is a single boost: the pure boost $\frac{\kappa}{2}B_1^+$, moved radially through the distance ℓ. The displacement is unique: it has no time component, its radial component is $-\ell$, and only translations along the axis itself remain free. Geometrically K is the meet of two hyperplanes, the observer's time hyperplane e_0 and the radial hyperplane moved to $e_1 - \ell e_4$:

$$K \;=\; \frac{\kappa}{2}\, e_0\,(e_1 - \ell\, e_4).$$

Its square $K^2 = \kappa^2/4$ does not see the translation. The Poincaré-invariant content of the static motion is the boost rate alone; the redshift only places the axis.

Where the axis is, and what it does, is read from events. Write the event with Minkowski coordinates x as $P(x) = (1 + e_4 x)\, i$. A translator moves events by its parameter, $T(\lambda)\,P(x)\,\widetilde{T}(\lambda) = P(x + \lambda)$, and a Poincaré bivector acts on them by

$$[\Omega^{(5)}_{\text{biv}}, P(x)] \;=\; e_4\left(\lambda + [\Omega_{\text{torsion}}, x]\right) i,$$

which is the velocity field $\lambda + \omega x$ of the Killing flow it generates. For K, along the radial line $x = s\,e_1$, the velocity is $(\nu + \kappa s)\,e_0$. It vanishes only at $s = -\ell$. The flow

$$\exp(\tau K) \;=\; T(-\ell e_1)\,R(\kappa\tau)\,\widetilde{T}(-\ell e_1)$$

fixes every event of the plane $\{t = 0,\ x = -\ell\}$ and carries the observer from the origin along

$$X(\tau) \;=\; \ell\big(\sinh\kappa\tau,\ \cosh\kappa\tau - 1,\ 0,\ 0\big),$$

a hyperbola that keeps the constant spacelike interval ℓ from the axis event $-\ell e_1$. Its radial coordinate is never negative: the orbit bends away from the axis. A hyperbola of interval ℓ has proper acceleration $1/\ell$ [19]. The acceleration of a Killing orbit is fixed by the Killing field and its first derivative at the event, so this is the true hover acceleration of the static observer, not only that of the flat flow.

Differentiating the hyperbola gives a tangent of squared interval $-(\ell\kappa)^2$. When $\nu = \ell\kappa > 0$, proper time along the orbit is $\sigma = \nu\tau$, and the rapidity at that proper time is σ/ℓ. In the inertial frame of the release, a body let go from rest travels on the straight line $s\,e_0$. The tangent is the unit timelike vector e_0, so s is proper time. The observer's motor in that frame is $\exp(\tau K)$. Seen from the observer, the inertial frame is the inverse motor

$$\exp(-\tau K) \;=\; T(-\ell e_1)\,R(-\kappa\tau)\,\widetilde{T}(-\ell e_1),$$

a boost of rapidity $-\kappa\tau$ about the same axis. The axis of the relative attitude is the axis of K at every instant.

The event of the straight line simultaneous with the observer is the event orthogonal to the unit velocity $(\cosh(\sigma/\ell),\ \sinh(\sigma/\ell))$. Its separation from the observer is the signed multiple

$$\xi(\sigma) \;=\; \ell\big(\operatorname{sech}(\sigma/\ell) - 1\big) \tag{14}$$

of the outward unit normal. For $\ell > 0$ the displacement is negative at every nonzero proper time: the body lies inward of the observer, on the side of the axis. At release, ξ and ξ' vanish, and

$$\xi''(0) \;=\; -\frac{1}{\ell}.$$

The same line meets the light cone of the axis event at proper time $s = \ell$, and that event lies on the Killing horizon $N = 0$ of the flat flow. The distance to the axis is therefore also the proper time in which a released body reaches that horizon. Along the observer's simultaneity the separation tends to $-\ell$ only as $\sigma \to \infty$: the body meets the horizon in finite proper time, while the one who hovers approaches the meeting asymptotically.

The same line is a law for radial rapidity. At proper time s with $|s| < \ell$, the body is passing the hovering observer whose axis lies at the hyperbolic distance $\rho = \sqrt{\ell^2 - s^2}$. Relative to that observer the radial rapidity ψ, taken positive outward, has $\sinh\psi = -s/\rho$. Along the body's proper time,

$$\frac{d\psi}{ds} \;=\; -\frac{\cosh\psi}{\rho}. \tag{15}$$

At release, $s = 0$, $\psi = 0$ and $\rho = \ell$, so the derivative is $-1/\ell$, the initial acceleration of the separation. Away from release the hyperbolic cosine remains in the law. A body already in motion changes its rapidity faster than the hovering acceleration, by the factor $\cosh\psi$, and the distance in the denominator is the local one. Even in the flat flow, ρ falls from ℓ to zero as the body approaches the light cone.

The direction of gravity at an event is the direction from the observer to the axis of its Killing bivector. Its strength is the inverse distance to that axis, the initial acceleration of a body released from rest.

On the exterior chart the distance and its inverse are

$$\ell \;=\; \frac{2r^2\sqrt{A}}{r_s}, \qquad a \;=\; \frac{1}{\ell} \;=\; \partial_r\sqrt{A} \;=\; \frac{r_s}{2r^2\sqrt{A}}.$$

The axis lies inward, and the initial acceleration of the flat flow there is $-a$. For a positive time translation the axis lies on the side opposite to the sign of κ, and κ has the sign of r_s: positive mass puts the axis toward the centre, and a negative r_s would put it outside and make the static field repel. The hover acceleration a has two further readings on the same chart. It is the component $T^0{}_{rt}$ of the Weitzenböck torsion (7): the torsion of the time leg is the gravitational force [20]. It is also the proper radial gradient of the rapidity, $a = -\sqrt{A}\,\varphi' = -d\varphi/ds$. Since $\varphi' < 0$, gravity points up the rapidity gradient, toward the region where the coframe is more strongly boosted, and ℓ is the proper length over which the rapidity would change by one unit at its present rate. Per unit proper time the generator is $K/\nu = \frac{1}{2}\big(a\,B_1^+ + N_0\big)$: a unit time translation together with a boost at the hover acceleration.

On the exterior the axis distance is no longer the hyperbolic radius of one Minkowski plane. It is the chart function $\ell(r)$. The rapidity law is kept, with that function in place of ρ, and the static coframe converts rapidity into the areal radius,

$$\frac{dr}{d\tau} \;=\; \sqrt{A}\,\sinh\psi, \qquad \frac{d\psi}{d\tau} \;=\; -\frac{\cosh\psi}{\ell(r)}. \tag{16}$$

The first relation is kinematics: ψ is the radial rapidity of the falling body in the orthonormal frame of the static observer, not the rapidity φ of the coframe. The second is the flat law, carried from event to event by the radius. Differentiate $\sqrt{A}\sinh\psi$. The radial derivative of $\sqrt{A}$ is $1/\ell$, so a changing redshift contributes $(\sqrt{A}/\ell)\sinh^2\psi$. The rapidity law contributes $-(\sqrt{A}/\ell)\cosh^2\psi$. Since $\cosh^2\psi - \sinh^2\psi = 1$, the sum is $-\sqrt{A}/\ell$, and every power of the speed cancels:

$$\frac{d^2r}{d\tau^2} \;=\; -\frac{\sqrt{A}}{\ell} \;=\; -\kappa \;=\; -\frac{r_s}{2r^2}. \tag{17}$$

The areal radius accelerates at the boost rate of the static Killing bivector, at every rapidity and not only at release. One further integration produces the radial Schwarzschild motion. The combination $E = \sqrt{A}\cosh\psi$ has vanishing derivative along the fall: redshift and rapidity trade exactly against each other. Therefore

$$\Big(\frac{dr}{d\tau}\Big)^2 \;=\; E^2 - A. \tag{18}$$

A body released from rest at radius R starts with $\psi = 0$, hence with $E = \sqrt{A(R)}$, and thereafter $\big(dr/d\tau\big)^2 = A(R) - A(r)$. The changing radius enters only by carrying $\ell(r)$ and $\sqrt{A}$ from event to event. What is integrated is the flat rapidity law. What comes out is the radial fall on the exterior chart.

The axis recedes strictly as r grows, and its two ends are the two limits of the exterior. Far away $\nu \to 1$, $\kappa \to 0$, $\ell \to \infty$, and K tends to the pure time translation $\frac{1}{2}N_0$: the axis is out of reach, the proper time to the horizon of the flat flow diverges, and there is no gravity. At the horizon $\nu \to 0$ and

$\ell \to 0$ while $\kappa \to 1/(2r_s)$: the axis reaches the observer, the proper time to the horizon of the flat flow vanishes, and K tends to the pure boost $\frac{1}{4r_s}B_1^+$, whose rate is the surface gravity of the horizon. The static motion is a translation at infinity and a boost at the horizon. In between it is a boost about an axis that stands at the finite distance $1/a$.

Inside a spherical mass the two parts of K separate. Gauss's law still fixes the boost rate by the enclosed mass, $\kappa = GM(< r)/r^2$, so κ vanishes at the centre with the weight. On the exterior the redshift is tied to that rate by $\kappa = d\nu/ds$. The continuum interior keeps the tie and integrates κ inward from the surface. At the centre, where $\kappa = 0$, K is a pure time translation at a reduced rate. For a uniform ball, in units $c = 1$,

$$\nu_{\text{cont}}(0) = \frac{3}{2}\sqrt{1 - \frac{2GM}{R}} - \frac{1}{2},$$

slower than the surface value $\sqrt{1 - 2GM/R}$ by $GM/(2R)$ in the weak field.

The parent theory does not keep the tie. A spherical shell cancels the boost parameters at every interior point, so an outer shell contributes neither mismatch nor redshift. The translation part is the exterior factor of the enclosed mass,

$$\nu(r) = \sqrt{1 - \frac{2GM(< r)}{r}}, \tag{19}$$

equal to 1 at the centre and to the exterior value on the surface. Weightlessness is the same pure time translation as at infinity: $\kappa = 0$ and $\nu = 1$, and K tends to $\frac{1}{2}N_0$. The axis recedes, as it does far away. The continuum centre is also weightless, and its axis likewise recedes, but the translation stays slow. That residual redshift is the potential the outer shells left behind. The equivalence principle as an identity equates the clock with the mismatch, and therefore equates a vanishing weight with $\nu = 1$.

Restoring c, with $GM = 3.986004418 \times 10^{14}\,\text{m}^3\,\text{s}^{-2}$ and $R = 6.371 \times 10^6\,\text{m}$, the centre clock runs fast relative to the surface by 6.96×10^{-10}, and the uniform continuum centre runs slow by 3.48×10^{-10}. Just below a crust of density $2.7 \times 10^3\,\text{kg}\,\text{m}^{-3}$ the fractional rate changes by $+5.12 \times 10^{-17}$ per metre on (19) and by -1.093×10^{-16} per metre in the continuum. Above the surface the two agree, both giving gh/c^2. The sign below the surface is the experiment that separates them. A few metres of depth lie within the reach of optical clocks. The centre is that comparison taken to the point where the weight is zero.

This also places the Killing bivector on the derivative ladder of Section 5. Its boost rate is a first derivative of the coframe, and its torsional scalar is

$$J(\kappa) \;=\; \tfrac{1}{2}\kappa^2 \;=\; A^2\,J_{\text{field}}, \qquad J(a) \;=\; \tfrac{1}{2}a^2 \;=\; A\,J_{\text{field}}.$$

Up to redshift, the field seed that fails to be $\frac{1}{2}T$ is the torsional scalar of the static observer's own boost.

Whether the Lorentz part is hyperbolic or elliptic decides on which side a body falls. Let $N^2(x) = -(\lambda + \omega x)^2$ be the squared Killing norm of the velocity field of an arbitrary Poincaré bivector. It is a quadratic function of the event, and its d'Alembertian is a constant,

$$\Box N^2 \;=\; 8J, \tag{20}$$

whatever the translation part and wherever it is evaluated. On the hyperbolic side the Killing norm grows away from the axis: along the radial line of K, $N^2 = (\nu + \kappa s)^2 = \kappa^2(s + \ell)^2$. On the elliptic side it shrinks: a rotation of rate ω in the (y, z)-plane, with a time translation ν along its axis, gives $N^2 = \nu^2 - \omega^2 y^2$, with $J = -\frac{1}{2}\omega^2$. An orbit of a Killing field accelerates along the gradient of $\log N$ [17]. Around a boost axis the orbit is pushed outward and free bodies fall in: the inertial force points to the axis. Around a rotation axis the orbit is pulled inward, centripetally, and free bodies drift out: the inertial force points away. Duality exchanges the two sides, since it sends B_1^+ to $-B_1^-$, the rotation of the (y, z)-plane, and reverses J. The rule of the parent theory, attraction for $J > 0$ and repulsion for $J < 0$, is in this reading the sign of the curvature of the Killing norm.

These statements concern the Killing jet at an event, the flat flow it generates, and the integral of that flow's rapidity law along the exterior. The straight line of one fixed jet is not itself the Schwarzschild geodesic. The geodesic is what the law becomes once ℓ and $\sqrt{A}$ are allowed to change with radius. No covariant transport of the full Killing bivector along a general curve is constructed, non-radial motion is not treated, and the torsional mismatch of the parent theory is not identified with a Killing bivector.

7 Conclusions

Adjoining a null vector e_4 to the biquaternion algebra of double spacetime realises translations as bivectors. The ten generators are a basis of the bivector space, and that space is the Poincaré algebra

$\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$: the six torsional generators close, and the four null generators span an abelian ideal on which they act. Duality $X \mapsto Xi$ is the complex structure of the Lorentz factor. Every boost is a dualised rotation, the bracket is complex-bilinear, and the sign that separates $\mathfrak{so}(3,1)$ from $\mathfrak{so}(4)$ is $i^2 = -1$. Translations are sent to grade four; duality does not close the null ideal.

The motor that unites the two sectors is the ordered product $M = RT$. Nilpotence of the null generator truncates its exponential to a translator $T = 1 + \Omega_{\text{trans}}$. The Lorentz factor has nilpotents of its own: a balanced off-axis sum of a boost and a rotation truncates to a null rotation of the Minkowski frame, an isometry whose parameter adds under multiplication and whose coefficients act unchanged on translations. Reversal-oddness of every bivector gives $M\widetilde{M} = 1$. When torsion and translation fail to commute, the product is not a single exponential; it remains the geometric object. Duality turns the torsional parameters through a quarter-turn, reversing J and preserving the unsigned mass. The diagnostic $J^{(5)}$ extends the Killing quadratic by the Minkowski pairing on translations; its translational term is unbounded off the light cone.

On a chart, sandwiching with M produces a tetrad. The exterior Schwarzschild coframe is a radial boost of that tetrad. The finite-angle scalar J, the field seed J_{field}, and the teleparallel density T are three different readings of the same geometry; they are not pointwise identical.

The direction of gravity is carried by the translations. The static observer's Killing bivector is a pure boost of rate $\kappa = GM/r^2$ about an axis moved inward through $\ell = 1/a$, where $a = \partial_r\sqrt{A}$ is the hover acceleration and the Weitzenböck torsion of the time leg. Gravity points to that axis. In the flat flow a body released from rest accelerates toward it at rate $1/\ell$ and meets its light cone in proper time ℓ. The same law, read for radial rapidity and integrated while ℓ changes with radius, is the radial fall on the exterior. The areal radius accelerates at the boost rate κ, and $(dr/d\tau)^2 = E^2 - A$, with $E = \sqrt{A}\cosh\psi$ constant along the motion. A release from rest at radius R has $E^2 = A(R)$. The axis recedes to infinity far away, where the static motion is a time translation, and reaches the observer at the horizon, where the motion is a boost at the surface gravity; the proper time ℓ diverges in the first limit and vanishes in the second. Up to redshift, the field seed is the torsional scalar of that boost. Because the squared Killing norm has d'Alembertian $8J$, a hyperbolic Lorentz part draws bodies toward its axis and an elliptic one drives them away; this is the kinematic content of the parent theory's sign rule. Inside a spherical mass the boost rate vanishes at the centre by the cancellation that removes the weight, and the translation returns to $\nu = 1$. A clock deep underground therefore runs fast relative to the surface, by 6.96×10^{-10} at the terrestrial centre and by 5.12×10^{-17} per metre just below a crust of density $2.7 \times 10^3\,\mathrm{kg\,m^{-3}}$. The continuum interior keeps $\kappa = d\nu/ds$ and is slowest where the weight is zero, by 3.48×10^{-10} at the centre of a uniform ball and by 1.093×10^{-16} per metre just below the surface. The sign of that comparison is the equivalence principle of the parent theory, read on K.

What remains open is a covariant bracket calculus for planes, a sandwich with the degenerate metric of $G(3,1,1)$, a field dictionary for a general motor, a transport of the full Killing bivector along a general curve, and the identification of that bivector with the mismatch of the parent theory. The radial rapidity on the exterior chart is the integral that has been carried out. The variational equivalence of TEGR with Einstein–Hilbert gravity is the input of [16].

A A plane bracket producing a translation

Planes in $G(3,1,1)$ are three-blades. For two three-blades A and B the antisymmetric and symmetric parts of the geometric product are

$$A \underline{\vee} B \;=\; \tfrac{1}{2}(AB - BA), \qquad A \,\overline{\vee}\, B \;=\; \tfrac{1}{2}(AB + BA).$$

These symbols are used only in this sense; they are not the anti-inner and anti-outer products of the PGA literature [2, 5]. In a five-dimensional vector space the geometric product of two grade-three elements has grades 0, 2, and 4 only.

Take $P = e_4 \wedge e_1 \wedge e_2$ and $D = jI = -e_0e_3e_2$, a dual-sector trivector of the parent theory. Anticommuting generators through the product $PD - DP$ reduces the bracket to a multiple of $e_4 \wedge e_3$. The direction e_3 is the unit normal to the spatial plane $e_1 \wedge e_2$, so

$$P \underline{\vee} D \;=\; k\,N_3$$

for a real coefficient k. The right-hand side is a generator of translations along that normal. The dual element D has acted as the carrier of the normal direction.

A second, purely spatial example confirms the grade pattern. For $A = e_1e_2e_3$ and $B = e_1e_2e_4$ one finds $AB = -e_3e_4$ and $BA = +e_3e_4$, hence $A \veebar B = -e_3e_4$ and $A \barwedge B = 0$: a pure grade-two commutator, as expected of an antisymmetric product.

B An extended reading

Sections 2–6 establish the algebra, the motor, the chart, and the axis, each with its own argument. The formulae are not repeated here. What that order does not pause to assemble is that three of its objects are one pair, that the scalars of the chart and the direction of the fall stand on one ladder, and that the sign which separates the Lorentz algebra from the Euclidean algebra is the sign which decides the side of a fall. This appendix is that assembly.

B.1 Orientation and displacement

The direct sum of Section 2 divides the bivector space into a Lorentz summand and a null ideal. Duality, in Section 3, is a complex structure on the first and an exit from grade two on the second. The motor has to be written so as to respect both. A map that organises boosts and rotations has already sent the translations out of the grade in which a rotor is an exponential, so it cannot be asked to organise displacement as well. The ordered product $M = RT$ is the form that keeps the two offices apart: R lies in the summand duality closes, and T lies in the ideal it leaves.

Reversal decides which grade that ideal may occupy, and the decision is the same one that makes a motor unitary. The cancellation $M\widetilde{M} = 1$ in Section 4 uses the reversal-odd character of every bivector together with the nilpotence of a null bivector. A grade-one null vector reverses evenly, and the cancellation stops. Realising translations as bivectors and requiring $M\widetilde{M} = 1$ are therefore one constraint on grade. Light-likeness remains a condition on the four coefficients, as Section 2 has it, so that the grade is available for every translation.

The complex structure has a second life on parameters. Complex bilinearity makes the boost–boost sign equal to $i^2 = -1$, and the quarter-turn $(\alpha, \beta) \mapsto (\beta, -\alpha)$ is that multiplication written on (α, β). Writing $z_a = \alpha_a + \mathrm{i}\beta_a$, the turn reverses the real part of $\sum_a z_a^2$ and preserves the Hermitian sum, which is the reversal of J and the preservation of the unsigned mass. The four coefficients λ^μ stand outside the turn. The Cartan–Killing form of the Poincaré algebra vanishes on the translation ideal [14, 15], so a quadratic that can see displacement has to bring in the Minkowski pairing by hand. The diagnostic $J^{(5)}$ is that admission: the complex structure measures orientation, and displacement is measured by a metric the structure does not see. Off the light cone the added term is unbounded, and any ceiling inherited from the discrete companion applies to the torsional parameters alone.

B.2 One pair, three writings

A radial boost set against a time translation appears three times. The appearances are one pair, written at three scales.

In Section 3 the pair is infinitesimal. The bracket $[B_0^+, N_0] = 2N_1$ witnesses that the Lorentz action on the null ideal is faithful. A change of orientation moves a translation. The ten-dimensional span is a semidirect product on the strength of that movement.

In Section 4 the pair is finite. When the factors commute, the exponential of the sum equals the ordered product RT. For this pair the two differ, and the geometric object is the product, which keeps an order. Conjugation says what the order does to a finite translation: every torsional rotor sends translators to translators, by its Lorentz action on λ^μ.

In Section 6 the pair is the static observer. The bivector K is the sum. The displacement identity says that this sum is already a pure boost moved through the distance $\ell = \nu/\kappa$. The finite flow $\exp(\tau K)$ is the conjugate of a boost of rapidity $\kappa\tau$, and so a product once more. The infinitesimal bracket, the disagreement of the exponential with RT, and the moved axis are the same failure to commute, read at a point, in a finite motor, and along an orbit.

The square $K^2 = \kappa^2/4$ does not register the translation that performed the move. The invariant content of the static motion is the boost rate. The redshift only places the axis. Strength and direction therefore separate inside one bivector. Strength is the Lorentz summand. Direction is the null summand, which duality could not keep in grade two. A rapidity is a scalar built from the Lorentz summand alone, and Section 5 can recover the exterior metric from it while still leaving the direction unsaid. The direction is said only when Section 6 restores the translation the chart did not use.

B.3 The ladder, and what points

Section 5 places three scalars on the exterior and keeps them distinct. The finite angle $J = \frac{1}{2}\varphi^2$ measures the radial boost. The field seed $J_{\text{field}} = \frac{1}{2}(\varphi')^2$ measures the square of its radial derivative. The teleparallel scalar T is a quadratic in the derivatives of the coframe. On the vacuum chart it equals $-2/r^2$, the value of the flat spherical frame, so it is not a function of the rapidity. A constant boost, a pure spatial translation, and the exterior point $(r_s, r) = (2, 4)$ keep the three rungs from collapsing into one another.

Section 6 climbs the same ladder with another bivector. The static observer is not the finite rotor of rapidity φ. It is the jet of ∂_t: the Lorentz part is the derivative $\kappa = \sqrt{A}\,\partial_r\sqrt{A}$, and the null part is the value $\nu = \sqrt{A}$. The torsional scalars of that jet, $\frac{1}{2}\kappa^2$ and $\frac{1}{2}a^2$, reproduce J_{field} up to the redshift factors A^2 and A. The field seed that disagrees with $\frac{1}{2}T$ agrees, up to redshift, with the angle-scalar of the observer's own boost. The disagreement and the agreement concern different rungs. T is a divergence built from the coframe. $J(\kappa)$ is the square of one component of its first derivative.

The inverse distance $1/\ell$ is that derivative under three names: the hover acceleration, the Weitzenböck component $T^0{}_{rt}$ of the time leg, and the proper gradient of φ. The chart already contained φ'. It acquires a direction when the unused translation νN_0 is restored and the ratio ν/κ is read as a displacement. For a positive mass the axis lies inward, and gravity points up the rapidity gradient, toward the more strongly boosted coframe. A scalar points only by being the rate of a boost about an axis that a translation has placed.

The two limits are the two ways that placement can fail. Far away the axis recedes, ℓ diverges, and K becomes a pure time translation: the Lorentz rung has vanished, and there is nothing to point toward. At the horizon the axis reaches the observer, ℓ vanishes, and K becomes a pure boost at the surface gravity: the null rung has vanished, and the direction has zero length. Between them the motion is a boost about an axis at the finite distance $1/a$.

A third limit sits at the centre of a spherical mass. The boost rate vanishes with the weight and the axis again recedes, while the translation returns to the asymptotic value. The continuum reaches the same recession by integrating κ inward, and arrives with a slowed translation. The buried clock of (19) runs fast relative to the surface. Far away and at the centre, K is a pure time translation at the vacuum rate. At the horizon it is a pure boost.

The fall of one jet is the flat flow. Its initial acceleration is $-1/\ell$, and the same ℓ is the proper time in which a body released from rest meets the light cone of the axis. The hovering observer approaches the meeting only as proper time tends to infinity. The acceleration of a Killing orbit is fixed by the jet, which is why $1/\ell$ is the hover acceleration on the exterior as well as in the flat model. Written for radial rapidity, that law may be integrated while ℓ changes with radius. The hyperbolic identity cancels the speed, the areal radius accelerates at the boost rate, and the first integral is $(dr/d\tau)^2 = E^2 - A$. The full bivector is still not carried along a general curve, and it is not identified with the torsional mismatch of the parent theory.

B.4 The sign is the complex structure

The parent theory distinguishes attraction and repulsion by the sign of J. Section 6 reads that sign as the curvature of the Killing norm, $\Box N^2 = 8J$. The reading has a source in Section 3 which the local computation does not require, and which shows the rule to be the complex structure evaluated on a flow.

The two squares of Section 2 are the two sides of the fall. Square $+1$ is the hyperbolic side: along the radial line of K the norm grows as one leaves the axis, the orbit is pushed outward, and a free body falls in. Square -1 is the elliptic side: a rotation in the (y, z)-plane makes the norm shrink as one leaves the axis, the orbit is drawn inward, and a free body drifts out. Duality exchanges the generators, sending B_1^+ to $-B_1^-$, and the quarter-turn exchanges the sign of J. The d'Alembertian changes sign with them. Attraction for $J > 0$ and repulsion for $J < 0$ are the boost–boost sign, the factor $i^2 = -1$, read on the flat flow of a single bivector. The sign that separates $\mathfrak{so}(3, 1)$ from $\mathfrak{so}(4)$ is the sign that chooses the side of the fall.

The translation does not enter $\Box N^2$. It places the axis and leaves the side to the Lorentz summand. This agrees with the split above. Duality, which reverses J, acts on the summand it closes, and the null ideal only locates the centre about which that curvature is felt.

B.5 Where the reading stops

Section 7 leaves four questions open. Each is the sentence that would follow the reading above, and each stops for a definite reason.

Appendix A shows that the antisymmetric product of one spatial plane with one dual trivector returns a null bivector along the normal. The configuration is a single one. A covariant bracket calculus would have to locate that product inside $\Omega^{(5)}_{\mathrm{biv}}$, so that the translation it returns is the translational half of a motor. The body records the grade and does not build the calculus.

The tetrad of Section 5 sandwiches the reference vectors of $\mathrm{Cl}(3,1)$. On the exterior chart the motor in that sandwich is a pure radial rotor, and the null direction which carries translations is not among the vectors being sent to the coframe. A sandwich taken in the degenerate metric of $G(3,1,1)$ is the step that would let the motor act in the algebra where the pointing direction lives. Until then, direction is read from the Killing bivector, while the tetrad carries the metric.

A dictionary for a general motor $M(x) = R(x)T(x)$ has to be built from derivatives of $(\Omega_{\mathrm{torsion}}, \lambda)$. The ladder already shows that the teleparallel scalar sees derivatives, and the static jet shows that the first derivative, written as a Poincaré bivector, is what points. The flat boost of an inertial frame is pure gauge: the metric stays Minkowski and the teleparallel scalar vanishes, while the torsion leg $\partial_t \varphi$ need not. The missing dictionary is the passage from a jet that changes the metric to a field of such jets. On the exterior chart T is a divergence, so that passage would be expected to contain a total divergence. The static jet remains the only metric-changing case written here.

The radial rapidity on the exterior is the integral of the flat law as the radius changes. To carry the bivector itself along a general manifold, to treat motion that is not radial, and to identify the bivector with the torsional mismatch of the parent theory, is a further construction. The variational equivalence of the teleparallel scalar with the Einstein–Hilbert action remains the input taken from [16].

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